GHANA COMMUNICATION TECHNOLOGY UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE

Course: Maths 105 Linear Algebra

PROGRAM: BSC. COMPUTER SCIENCE (CYBERSECURITY)

LEVEL: L100

Name:

Index No:

ASSIGNMENT no 1

Question: How Can We Apply Linear Algebra in Cybersecurity?

Cryptography:
Subject example: Hill Cipher (1929)

The Hill Cipher is the first encryption technique which crossed us over encryption as linguistics to encryption as mathematics and it utilized elements of linear algebra. Even though it's a weak one compared to advanced algorithms such as Advanced Encryption Standards, RSA etc. But in the light of this, it's worth the subject.

Letter–Number Mapping: Moving into a Finite Field

Cryptography begins by converting language into mathematics.

Each letter is mapped to a number:

A → 0

B → 1

Z → 25

This creates an alphabet inside:

$F_{26} = \{0,1,2,\ldots,25\}$

All arithmetic is done **mod 26**.

For example:

$27 \equiv 1 \pmod{26}$

It transforms text into vectors inside a **finite vector space**.

A block of 3 letters like:

"CAT"

becomes a vector:

[\matrix(2@0@19)]

Now the message is no longer linguistic.

It is an element of:

[

(\mathbb{F}_{26})^3

]

That is a vector space.

Linear algebra work here:

Messages are now vectors over a finite field.

Matrix Multiplication in the Hill Cipher

The Hill Cipher applies a matrix transformation to message blocks.

Let the encryption matrix be:

[

E =

\begin{bmatrix}

2 & 3 & 1 \

1 & 1 & 0 \

4 & 0 & 1

\end{bmatrix}

\mod 26

]

Encryption is defined as:

[

C = E v \mod 26

]

Where:

- (v) = plaintext vector
- (C) = ciphertext vector

This is pure matrix multiplication.

Each ciphertext letter becomes a **linear combination** of plaintext letters.

For example:

[

c_1 = 2v_1 + 3v_2 + 1v_3 \ (\text{mod } 26)

]

This is exactly how linear transformations work in linear algebra:

[

T(v) = Ev

]

Decryption requires:

[

v = E^{-1} C \mod 26

]

So encryption and decryption rely on:

- Determinants
- Matrix invertibility
- Linear independence
- Modular arithmetic

All core concepts of linear algebra.